

Tutorial Sheet 4

Introduction to Numerical Methods for ODEs and PDEs

Partial Differential Equations (PDEs)

Masoud Seifkar

Autumn Term 2023

Department of Aeronautics, Imperial College London

Part I: Practical Exercises

Exercise 1 (Wave equation)

Consider the 1st order wave equation, which is described by the partial differential equation (PDE)

$$\frac{\partial u}{\partial t} + c \frac{\partial u}{\partial x} = 0,$$

for $0 \leq x \leq L$ and $t \geq 0$ and with boundary conditions $u(0, t) = 0$ and

$$u(x, 0) = \exp(-200(x - 0.2)^2).$$

Let $c = 1$ and $L = 1$.

$$\begin{aligned} \frac{\partial u}{\partial t} &= -c \frac{\partial u}{\partial x} = - \frac{\partial u}{\partial x} \\ \frac{\partial u}{\partial x} &= \frac{u_{j+1} + 0u_j - u_{j-1}}{2\Delta x} \text{ where } j = 1, \dots, N-1 \\ \therefore \frac{\partial \bar{u}}{\partial t} &= -c \left(\frac{u_{j+1} + 0u_j - u_{j-1}}{2\Delta x} \right) \end{aligned}$$

1. By performing a semi-discretisation with a uniform spacing Δx , using 2nd order central differencing, show that you can approximate the PDE by a system of ordinary differential equations (ODEs) of the form

$$\frac{d\bar{u}}{dt} = -\frac{c}{2\Delta x} B[-1, 0, 1] \bar{u}$$

where $\bar{u}$ is a vector of the values of $u(x, t)$ at each point in the spatial discretisation, i.e. $\bar{u} = [u_1, u_2, \dots, u_{N-1}]^T$ and $B[\cdot, \cdot, \cdot]$ denotes a banded (tridiagonal) matrix.

2. What is the initial condition of the ODE obtained in part 1 of this exercise?
3. Let $\Delta x = 0.01$. Using a time step $\Delta t = 0.01$ (Δt is what we denoted as h in the context of ODEs), use explicit Euler to obtain a numerical solution of the PDE.
4. Let $\Delta x = 0.05$. Using a time step $\Delta t = 0.01$, use explicit Euler to obtain a numerical solution of the PDE. Comment on the result with respect to the numerical solution obtained in part 3 of this question.
5. Let $\Delta x = 0.002$. Try to obtain a numerical solution using explicit Euler for the time-advancement scheme, using time step a time step $\Delta t = 0.01$. What do you observe?
6. Let $\Delta x = 0.002$. Obtain a numerical solution using implicit Euler for the time-advancement scheme, using time step a time step $\Delta t = 0.005$. Comment on what you observe

Exercise 2

The equation describing the motion of a guitar string is given by

$$\frac{\partial^2 u}{\partial t^2} - c^2 \frac{\partial^2 u}{\partial x^2} = 0,$$

where $0 \leq x \leq L$ denotes the spatial variable, $L = 1$ is the length of the string, $c > 0$ and t denotes the temporal variable. The boundary conditions are $u(0, t) = u(L, t) = 0$ (since the string is fixed at both ends) and the initial conditions are $u(x, 0) = \sin(\pi x)$ (note that this initial condition respects the boundary conditions, *i.e.* $u(0, 0) = u(L, 0) = 0$). The solution $u(x, t)$ represents the vertical displacement of the string.

1. Find an approximation for $\frac{\partial^2 u}{\partial t^2}$ by performing a semi-discretisation of the spatial derivative.
2. The result of part 1 of this question is a second order ODE. Convert it into a (larger) system of first order ODEs.
3. Obtain a solution for the first order ODE obtained in part 2, using 4th order Runge-Kutta (experiment with and select the time step and spatial grid spacing as you see fit).

$$\begin{aligned} 1. \quad \frac{\partial^2 u}{\partial t^2} &= c^2 \frac{\partial^2 u}{\partial x^2} \\ \frac{\partial^2 u}{\partial x^2} &= \frac{u_{j+1} - 2u_j + u_{j-1}}{\Delta x^2} \quad \text{where } j = 1, \dots, N-1 \\ \therefore \frac{\partial^2 \bar{u}}{\partial t^2} &= c^2 \left(\frac{u_{j+1} - 2u_j + u_{j-1}}{\Delta x^2} \right) \\ &= \frac{c^2}{\Delta x^2} \begin{bmatrix} 1 & -2 & 1 \end{bmatrix} \bar{u} \quad \text{for } \bar{u} = \begin{bmatrix} \overset{=0}{u_1} & u_2 & \dots & u_{N-1} & \overset{=0}{u_N} \end{bmatrix}^T \end{aligned}$$

$$2. \quad u_1 = u, \quad u_2 = \frac{du}{dt} \quad \therefore u_1' = u_2, \quad u_2' = \frac{du}{dt} = \frac{c^2}{\Delta x^2} \begin{bmatrix} 1 & -2 & 1 \end{bmatrix} u_1$$

$$\text{At } t=0, \quad u_1(0) = u(x, 0), \quad u_2(0) = \frac{du}{dt}(x, 0) = 0$$

$$\therefore \frac{d\bar{u}}{dt} = \begin{bmatrix} u_2 \\ \frac{c^2}{\Delta x^2} \begin{bmatrix} 1 & -2 & 1 \end{bmatrix} u_1 \end{bmatrix} \quad \text{where } u_1(0) = \begin{bmatrix} \sin(\pi x) \\ 0 \end{bmatrix}$$

Part II: Theoretical Considerations

Exercise 3

Consider the heat equation

$$\frac{\partial u}{\partial t} = \alpha \frac{\partial^2 u}{\partial x^2},$$

and its numerical solution using second order spatial differencing and explicit Euler for its time-advancement scheme.

1. Derive the modified equation corresponding to this numerical scheme
2. Show that the accuracy of the scheme can be improved by selecting the spatial grid spacing Δx and the time step Δt such that

$$\frac{\alpha \Delta t}{\Delta x^2} = \frac{1}{6}.$$

Exercise 4

Consider the heat equation

$$\frac{\partial \phi}{\partial t} = \alpha \frac{\partial^2 \phi}{\partial x^2}.$$

1. Derive the ODE which approximates the heat equation if you perform a semi-discretization of the spatial variable x using second order central differencing.
2. Suppose you are to solve the system of ODEs obtained in part 1 of this question using the trapezoidal method. Using modified wavenumber analysis, show that the resulting numerical scheme is unconditionally stable.

See pre-lecture notes 6